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Automated waste classification for smart recycling: A multi-class CNN approach with transfer learning and pre-trained models

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ABSTRACT

This study presents a comparative analysis of pre-trained CNN models using transfer learning for the multi-class classification of recyclable waste. The dataset used is publicly available and comprises 15,150 images belonging to 12 different categories of recyclable household waste: paper, cardboard, biological, metal, plastic, green-glass, brown-glass, white-glass, clothes, shoes, batteries, and trash. The methodology involves implementing 14 different CNN models pre-trained on ImageNet (e.g., ConvNeXt_V1, ResNet50, MobileNet_V3_Large) on this dataset through a transfer learning procedure. In all pre-trained models, only the final classifier layer was trained, while all other layers were kept frozen, utilizing a feature extraction strategy that leverages previously learned visual representations. This approach significantly reduced training time and computational cost, while minimizing the risk of overfitting in scenarios with limited data. Model performance was evaluated using classification metrics such as accuracy, F1-score, precision, and recall, as well as computational efficiency indicators including architectural depth, parameter count, and training time. According to the findings, the ConvNeXt_V1 model achieved the highest performance with a test accuracy of 96.95 %. On the other hand, MobileNet_V2 (93.90 %), MobileNet_V3_Large (93.69 %), MNASNet_1_0 (93.69 %), RegNet_Y_800MF (92.87 %), and SqueezeNet_1_0 (92.10 %) offered shorter training times (ranging from 2856 to 7219 s) and stood out as efficient alternatives for embedded system applications.

1. Introduction

The rapid global increase in population, urbanization, and consumption habits has led to a dramatic rise in waste generation, making it one of the fundamental challenges threatening environmental sustainability. The increasing amount of municipal, industrial, and commercial waste depletes natural resources, disrupts ecosystems, and raises greenhouse gas emissions. (Chen et al., 2020; Sahoo et al., 2024). In this context, it is critically important not only to dispose of waste in an environmentally responsible manner but also to recycle it and reintegrate it into the economy whenever possible (Adami and Schiavon, 2021; Arbeláez-Estrada et al., 2023). The effectiveness of recycling processes plays a vital role not only in generating economic value but also in reducing natural resource consumption, conserving energy, and minimizing carbon footprints (Chen et al., 2020; Adami and Schiavon, 2021; Arbeláez-Estrada

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et al., 2023; Gurcan, 2024; Sahoo et al., 2024).

Most current systems still rely on manual or semi-automated waste sorting methods, which are inefficient and error-prone (Gurcan, 2023; Islam et al., 2025b; Konstantinidis et al., 2023; Pitakaso et al., 2024; Zhang et al., 2021). These processes tend to be inefficient due to high labor requirements, low operational speed, and susceptibility to human error. This in turn limits recycling rates and leads to quality losses, especially in heterogeneous waste streams (Arbeláez-Estrada et al., 2023; Konstantinidis et al., 2023; Kumar et al., 2021). Automation of waste classification is crucial for both environmental sustainability and the efficiency of recycling chains (Adami and Schiavon, 2021; Arbeláez-Estrada et al., 2023; Nahiduzzaman et al., 2025).

Recently, computer vision and deep learning have become key tools for automated waste classification. (Islam et al., 2025b; Nahiduzzaman et al., 2025; Xie et al., 2025; Malik et al., 2022). In particular, convolutional neural networks (CNN) have proven to be effective in distinguishing between different types of waste due to their high accuracy and strong learning capabilities in image classification tasks (Bobulski and Kubanek, 2021; Altikat et al., 2022; Li and Chen, 2023). These systems offer both high classification performance and reduced dependence on manual labor. Deep learning-based models can achieve successful classification even when traditional methods fail, particularly for waste categories with high visual similarity and intra-class variation (Altikat et al., 2022; Yevle and Mann, 2025; Nahiduzzaman et al., 2025; Islam et al., 2025b). Unlike conventional approaches, these models utilize data-driven learning to directly classify waste from images, thereby accelerating the process and reducing errors (Yong et al., 2023; Wu et al., 2023; Arbeláez-Estrada et al., 2023; Fotovvatikhah et al., 2025).

Deep CNNs often require extensive datasets, long training times, and high computational power (Wu et al., 2023; Yevle and Mann, 2025; Fotovvatikhah et al., 2025). These requirements pose significant challenges, particularly for applications deployed in mobile or embedded systems. Transfer learning reuses pre-trained models, lowering training costs and maintaining high performance with limited data (Zhang et al., 2021; Yong et al., 2023; Lilhore et al., 2024; Oza et al., 2025). Specifically, feature extraction–based frozen transfer learning strategies, in which only the final classification layer is retrained while the rest of the network remains frozen, help minimize computational load and the risk of overfitting. This approach offers notable advantages for scenarios like waste classification, where access to large datasets may be limited (Huang et al., 2020; Zhang et al., 2021; Lilhore et al., 2024; Islam et al., 2025b).

Previous studies have shown that CNN architectures used in waste classification problems vary significantly in structural aspects such as filter size, depth, connectivity patterns, and parameter density (Wu et al., 2023; Arbeláez-Estrada et al., 2023; Yevle and Mann, 2025; Fotovvatikhah et al., 2025; Yasin and Koklu, 2025). These CNN models have evolved to meet requirements in data scale, class balance, and hardware constraints, inspiring new optimization strategies (Islam et al., 2025b; Wu et al., 2023; Malik et al., 2022). Early works primarily employed conventional architectures like ResNet, VGG16, and Inception for basic classification tasks; however, their large parameter sizes and high computational costs posed constraints for resource-limited systems (Huang et al., 2020; Bian et al., 2021; Bobulski and Kubanek, 2021; Faria et al., 2021; Zhang et al., 2021; Komathi et al., 2023). Subsequently, efforts shifted toward the development of lightweight and mobile-friendly networks, such as MobileNet series, MNASNet, and EfficientNet, which aim to balance accuracy and efficiency through architectural optimization and modular or automated design (Yong et al., 2023; Komathi et al., 2023; Malik et al., 2022; Poudel and Poudyal, 2022). Recently, hybrid architectures combining Transformer-based attention with convolutional frameworks, such as ViT, ConvNeXt, and Swin Transformer, have gained attention for improving contextual discrimination in similar classes (Prova, 2024; Madhavi et al., 2025; Islam et al., 2025a, b; Oza et al., 2025; Nahiduzzaman et al., 2025). Despite these architectural advancements, most existing studies tend to focus solely on overall classification accuracy (Wu et al., 2023; Yevle and Mann, 2025; Fotovvatikhah et al., 2025). Comparative analyses that evaluate CNN models in terms of training time, parameter efficiency, per-class performance, and visual confusion trends remain limited in the literature (Arbeláez-Estrada et al., 2023; Islam et al., 2025b; Nahiduzzaman et al., 2025; Malik et al., 2022).

This study aims to address this gap by conducting a comprehensive evaluation of CNN architectures based on transfer learning using multidimensional criteria for the task of waste classification. To this end, 14 widely used pre-trained CNN models (e.g., VGG16, ResNet50, ConvNeXt_V1, MobileNetV3), each with distinct structural characteristics, are systematically compared under the same dataset, training protocol, and evaluation metrics. All models employ a feature extraction–based frozen transfer learning approach, in which only the final classifier layer is trained. This strategy significantly reduces training time and computational cost while minimizing overfitting risks in data-constrained settings. Accordingly, this study makes the following original contributions to the literature:

- It analyzes 14 pre-trained CNN architectures in terms of design intent, architectural type, and release year, highlighting the evolution of CNN model design.
- It evaluates computational efficiency based on architectural depth, parameter count, model size, and CPU training time.
- All models use a feature extraction–based frozen transfer learning strategy, where only the final layer is trained, minimizing training time and resource use while reducing overfitting under limited data conditions.
- Models are trained under a unified dataset and protocol, enabling fair and systematic comparison of their overall classification performance in the waste classification task.
- Per-class F1-score analyses reveal the strengths and weaknesses of each model in classifying different waste categories, accounting for factors such as visual similarity, sample imbalance, and intra-class diversity.
- Model performances are visualized using heatmaps based on confusion matrices and precision-recall-F1 metrics, facilitating the identification of high-confusion class pairs.
- Models that achieve a balance between high accuracy and low resource consumption are identified, offering architecture recommendations suitable for real-world and embedded system deployments.
- Unlike most existing studies, this work presents an integrated evaluation framework that jointly considers model accuracy, class-level performance, and computational efficiency.

2. Background and related work

In recent years, deep learning-based methods have gained significant momentum in waste classification systems, particularly due to the high classification performance and automatic feature extraction capabilities of image-based models (Yevle and Mann, 2025; Fotovvatikhah et al., 2025). Early studies commonly utilized CNN architectures trained from scratch. Although these models yielded effective results in tasks with a limited number of classes, they presented limitations in terms of generalizability and training cost when applied to large and visually complex datasets (Huang et al., 2020; Fotovvatikhah et al., 2025). During 2020–2021, widely used models such as VGG16, ResNet50, and DenseNet achieved accuracy rates ranging from 85 % to 96 % on datasets with 4–6 classes (Faria et al., 2021; Zhang et al., 2021; Komathi et al., 2023). However, their high parameter counts, long training times, and tendency to overfit make these models unsuitable for resource-constrained systems, limiting their use in practical applications (Huang et al., 2020; Faria et al., 2021; Zhang et al., 2021; Arbeláez-Estrada et al., 2023). As a result, recent research has shifted toward transfer learning (TL) strategies, which aim to reuse CNN models pre-trained on large datasets for different downstream tasks (Komathi et al., 2023; Yong et al., 2023; Mishra et al., 2024; Yevle and Mann, 2025).

Transfer learning-based studies have enabled higher accuracy and shorter training times in waste classification tasks involving limited data. For example, Faria et al. (2021) reported 88.42 % accuracy using the VGG16 model in a four-class task, while Komathi et al. (2023) achieved 92.6 % accuracy with ResNet50 and Inception on a six-class dataset (Faria et al., 2021; Komathi et al., 2023). In another study conducted by Malik et al. (2022), EfficientNet-B0 achieved 85 % accuracy in a ten-class classification task (Malik et al., 2022). These results demonstrate that transfer learning architectures can generalize effectively even in datasets with limited samples and high visual diversity (Zhang et al., 2021; Poudel and Poudyal, 2022; Oza et al., 2025). In contrast, studies involving models trained from scratch tend to be narrower in scope and more data-intensive (Wu et al., 2023; Fotovvatikhah et al., 2025). For instance, Bobulski and Kubanek (2021) achieved 96.41 % accuracy across four classes using a custom-designed CNN model (Bobulski and Kubanek, 2021). Similarly, Chhabra et al. (2024) reported 93.28 % accuracy using a multi-layer deep architecture, albeit limited to a binary classification task (Chhabra et al., 2024). While these models offer strong performance, they are often associated with high architectural complexity and long training times, making them less feasible for practical implementation due to increased data requirements (Jain et al., 2022; Yevle and Mann, 2025; Fotovvatikhah et al., 2025).

Hybrid approaches involve classifying features extracted via CNNs using traditional machine learning algorithms such as SVM, ELM, and GBT. Mishra et al. (2024) reached 99.84 % accuracy by combining DenseNet121 features with an SVM classifier (Mishra et al., 2024). Al Duhayyim et al. (2023) reported 98.85 % accuracy in a binary task using a fusion of SVM, ELM, and GBT classifiers (Al Duhayyim et al., 2023). While such hybrid systems offer high accuracy, their structural complexity and latency sensitivity pose challenges for real-time deployment.

As of 2022, new architectural trends focusing on balancing accuracy and efficiency have become more prominent in the literature. Lightweight NAS-based architectures such as MobileNet, EfficientNet, and MNASNet have been favored for their low hardware requirements and reduced training times (Yong et al., 2023; Malik et al., 2022). hybrid models that integrate Transformer mechanisms into convolutional frameworks, such as ViT, Swin Transformer, and ConvNeXt, have achieved high accuracy in multi-class classification tasks. Huang et al. (2021) obtained 96.48 % accuracy using a Vision Transformer (ViT) on a six-class dataset, while Prova (2024) reported 95 % accuracy in a 12-class setting with a ViT-based system (Huang et al., 2021; Prova, 2024). Moreover, Transformer-inspired CNNs such as ConvNeXt aim to combine classical convolutional architectures with modern optimization techniques to improve both accuracy and generalizability (Liu et al., 2022). Madhavi et al. (2025) reported one of the highest accuracies in the literature (98.97 %) using a model combining Swin Transformer and ConvNeXt architectures (Madhavi et al., 2025).

Despite these advancements, most studies in the literature have not explored per-class performance, misclassification trends due to visual similarity, or the impact of architectural design on computational load within a comprehensive framework. To address this gap, the present study systematically compares 14 different pre-trained CNN architectures under equal conditions; analyzes class-level accuracy, precision, recall, and F1-score to identify challenging categories; and offers model recommendations tailored to real-world application scenarios based on parameter efficiency and training time metrics.

3. Materials and methods

The methodology employed in this study utilizes advanced deep learning and transfer learning techniques for effective waste classification. In particular, it is based on the use of pre-trained deep CNN models. The aim of the study is to accurately classify waste images using the publicly available Waste Classification dataset, which contains images from 12 different waste categories. The overall methodology is illustrated in Fig. 1, which comprehensively represents the architecture and workflow used in the study. The figure sequentially demonstrates how deep learning techniques and transfer learning are integrated, as well as how pre-trained CNN models are employed throughout the process.

The proposed architecture begins with the data preparation phase. At this stage, waste images are collected, subjected to preprocessing steps, and made ready for classification. Then, 14 different pre-trained CNN models, originally trained on the ImageNet dataset and adapted for transfer learning, are employed. These models automatically extract meaningful visual features from the images, and based on these features, the waste samples are assigned to their corresponding categories. In the final stage, model performance is evaluated using various metrics, including accuracy, precision, recall, F1-score, and the confusion matrix.

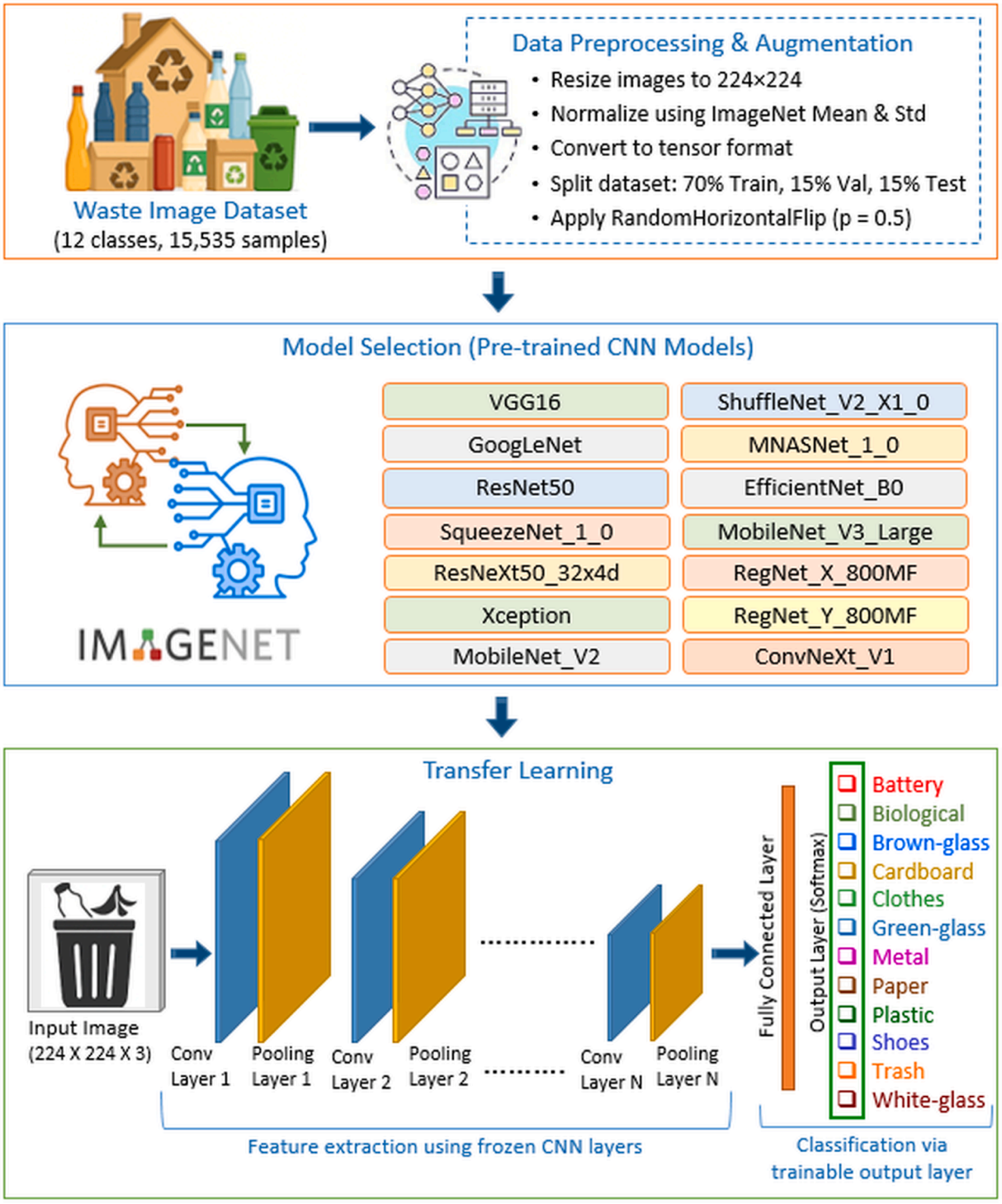


Fig. 1. Workflow and process representation of the proposed methodology.

3.1. Data description

The dataset used in this study is the publicly available Waste Classification dataset, hosted on the Kaggle platform (Kaggle, 2025). It contains a total of 15,535 images across 12 distinct waste categories. The class distribution is as follows: clothes (5325), shoes (1977),

paper (1050), biological (985), battery (945), cardboard (891), plastic (865), white-glass (775), metal (769), trash (697), green-glass (629), and brown-glass (607) (Fig. 2). This distribution reveals a notable imbalance in the number of samples across classes. The clothes, shoes, trash, and biological classes exhibit high visual variability in terms of shape, texture, content, and form. For instance, the clothes class includes subtypes such as jackets, pants, shirts, and dresses, while the shoes class covers various footwear types such as sneakers, boots, and sandals. The trash and biological categories are composed of heterogeneous waste materials such as organic residue, packaging, and mixed refuse. In contrast, classes such as brown-glass, green-glass, metal, and plastic are more visually homogeneous due to limited color diversity and similar material or surface properties (e.g., glossy textures of glass or metallic finishes) (Prova, 2024; Madhavi et al., 2025).

However, it would be misleading to assume that differences in sample counts across classes are solely due to visual diversity. This distribution may also have been influenced by random factors or source-based sampling preferences during the dataset's construction. Therefore, in order to ensure the reproducibility of the study and maintain experimental consistency, the dataset was included in the analysis in its original form, without any modifications to the number of classes or the sample distributions. Accordingly, the modeling and evaluation procedures were designed by taking into account both the sample imbalance across classes and the differences in visual structure. Fig. 3 illustrates the diversity of the dataset and the visual differences between classes by presenting three randomly selected sample images from each category.

3.2. Data preprocessing and augmentation

In order to improve the learning process of the model, reduce overfitting, and enhance classification performance, various preprocessing and data augmentation techniques were applied to the dataset prior to training. The pre-trained models used in this study were trained on the ImageNet dataset, and therefore have learned not only low-level visual features such as edges, colors, and textures, but also high-level conceptual representations required for object recognition. In this way, these models can be transferred to smaller or different datasets, making it possible to achieve higher accuracy and a faster training process with limited data (Abou Baker et al., 2022; Disci et al., 2025).

To match the input expectations of the models, all images in the dataset were resized to 224 × 224 pixels, converted to tensor format, and scaled to pixel values in the range of 0–1. They were then normalized using the mean and standard deviation values of the RGB color channels: 0.485, 0.456, 0.406 (mean) and 0.229, 0.224, 0.225 (standard deviation), respectively. These values are statistical parameters computed over the ImageNet dataset. Thus, by preserving the input distribution expected by the models, the efficiency of transfer learning was improved (Jain et al., 2022; Prova, 2024; Wu et al., 2024).

As part of data augmentation, a RandomHorizontalFlip operation was applied, horizontally flipping each image with a 50 % probability, thereby increasing the model's robustness against directional variation. Due to significant imbalance in the number of samples across classes in the dataset, class weights were calculated based on the number of samples per class and integrated into the CrossEntropyLoss function. In this way, minority classes were given more representation during training, enabling the model to learn all classes more evenly. The intensity of data augmentation was intentionally kept minimal to avoid excessive artificial variations in small datasets (Cui et al., 2019).

The dataset was randomly but reproducibly split into three subsets using a fixed random seed: 70 % for training, 15 % for validation, and 15 % for testing. These ratios ensured that the model was trained on a sufficient number of samples, while also allowing for

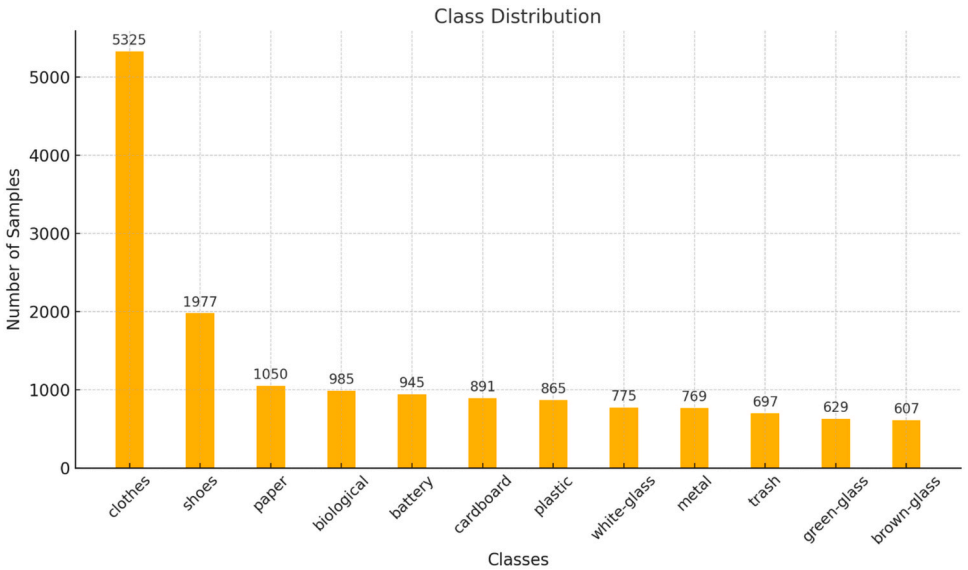


Fig. 2. Class distribution of the dataset.



Fig. 3. Three sample images from each class in the dataset.

an unbiased evaluation of overall performance through the validation and test sets. In addition, the validation set was used to monitor the model's tendency to overfit, while the test set, never seen by the model during training, was used to evaluate its generalization performance. A fixed random seed was used in data splitting and loading operations to ensure deterministic reproducibility, while multiple randomized splits or k-fold validation could further assess the statistical robustness of the models.

3.3. Implementation of pre-trained CNN models

In this study, a total of 14 pre-trained models with different CNN architectures were employed. Within the framework of transfer learning, these models were adapted to the new task of waste classification by leveraging prior knowledge, rather than training from scratch, enabling faster and more efficient learning. The pre-trained CNN models used are as follows: MobileNet_V2, MobileNet_V3_Large, ResNet50, ResNeXt50_32x4d, EfficientNet_B0, ConvNeXt_V1, RegNet_X_800MF, RegNet_Y_800MF, GoogLeNet, ShuffleNet_V2_X1_0, Xception, MNASNet_1_0, SqueezeNet_1_0, and VGG16 (Zhang et al., 2021; Abou Baker et al., 2022; Kaya and Gürsoy, 2023; Yevle and Mann, 2025). Table 1 compares these models in terms of architectural category, design type, intended goal, and release year, allowing for a joint evaluation of lightweight architectures with low resource consumption and deep models providing high accuracy.

The models belong to various architectural categories such as plain CNN, inception-based, residual, aggregated, lightweight, depthwise-separable, NAS-based, design-optimized, SE-enhanced, and transformer-inspired convolutional networks. Early architectures like VGG16 are based on stacked convolutional layers, whereas more advanced ones like ResNet50 and ResNeXt50_32x4d incorporate residual and grouped convolutions to prevent gradient vanishing in deep networks (Khan et al., 2020; Abou Baker et al., 2022). Lightweight models such as MobileNet_V2, SqueezeNet_1_0, ShuffleNet_V2_X1_0, and MNASNet_1_0 are optimized for mobile and embedded systems due to their reduced parameter counts and computational cost (Khan et al., 2020; Yong et al., 2023; Arbeláez-Estrada et al., 2023; Fotovvatikhah et al., 2025). These architectures utilize efficient building blocks such as depthwise convolutions, inverted residuals, and channel shuffle operations. Notably, MNASNet_1_0 and MobileNet_V3_Large were automatically constructed via Neural Architecture Search (NAS) to achieve optimal accuracy-efficiency tradeoffs (Khan et al., 2020; Zhang et al., 2021; Oza et al., 2025).

EfficientNet_B0 employs a compound scaling strategy to jointly optimize depth, width, and input resolution, maximizing performance under resource constraints. RegNet_X and RegNet_Y families were developed through design space exploration, with the Y variant enhanced by squeeze-and-excitation (SE) blocks to increase representational capacity (Radosavovic et al., 2020; Malik et al., 2022). ConvNeXt_V1 represents a modern architectural approach that enhances convolutional networks with Transformer-inspired components such as large kernels, GELU activation, LayerNorm, and decoupled weight decay (Liu et al., 2022; Madhavi et al., 2025).

All models were pre-trained on ImageNet, a widely used benchmark dataset containing approximately 1.2 million labeled images across 1000 object categories (Russakovsky et al., 2015; Abou Baker et al., 2022). Training on ImageNet enables these models to learn general visual representations such as edges, colors, and textures, which can be effectively transferred to new tasks with limited data and computation. In each model, the original fully connected classification head was removed and replaced with a task-specific output layer for 12 waste categories (Jain et al., 2022; Pitakaso et al., 2024). All layers except the new classifier were frozen, and only the final layer was updated during training via backpropagation. This strategy, known as feature extraction-based transfer learning, preserves the convolutional feature extraction layers pre-optimized on large datasets and retrains only the classification head. As a result, training time and computational load are significantly reduced, and the risk of overfitting is minimized (Zhang et al., 2021; Abou Baker et al., 2022).

Architectural and parametric characteristics of the models are presented in Table 2. This table compares structural depth, total and trainable parameters, and model size. The metrics were obtained through experimental analysis based on the official TorchVision implementations of the ImageNet pre-trained versions (Marcel and Rodriguez, 2010). The Total Params column shows all weights in the network, while the Trainable Params reflect only the parameters updated during training. Since only the classification head was retrained, trainable parameters constitute a small portion of the total. For instance, while VGG16 has 134 million parameters in total, only 49.2 K were trainable, highlighting the efficiency of the frozen transfer learning strategy.

3.4. Model compilation and evaluation

To ensure a fair and reproducible comparison, all 14 pre-trained CNN architectures were trained under a unified experimental

Table 1
Summary of the CNN models by year, category, architecture type, and design goal.

Model	Year	Category	Architecture Type	Design Goal
VGG16	2014	Plain CNN	Stacked Convolutions	High Accuracy
GoogLeNet	2014	Inception CNN	Inception Modules	Balanced Efficiency
ResNet50	2015	Residual CNN	Residual Connections	High Accuracy
SqueezeNet_1_0	2016	Lightweight CNN	Fire Modules (1 ×1 & 3 ×3)	Mobile Optimization
ResNeXt50_32x4d	2017	Aggregated CNN	Grouped Convolutions	Balanced Efficiency
Xception	2017	Depthwise CNN	Depthwise Separable Convolutions	High Accuracy
MobileNet_V2	2018	Lightweight CNN	Depthwise + Linear Bottleneck	Mobile Optimization
ShuffleNet_V2_X1_0	2018	Lightweight CNN	Grouped Conv + Channel Shuffle	Mobile Optimization
MNASNet_1_0	2018	NAS-Based CNN	Neural Architecture Search	Mobile Optimization
EfficientNet_B0	2019	NAS + Scaled CNN	Compound Scaling (Width/Depth/Res)	Balanced Efficiency
MobileNet_V3_Large	2019	NAS + Lightweight CNN	NAS + Depthwise Separable	Mobile Optimization
RegNet_X_800MF	2020	Design-Search CNN	Automated Design Space Search	Balanced Efficiency
RegNet_Y_800MF	2020	SE-Enhanced CNN	Design Search + Squeeze-Excitation	Balanced Efficiency
ConvNeXt_V1	2022	Transformer CNN	ConvNet w/ Transformer Tricks	High Accuracy

Table 2
Summary of the CNN models with architectural depth and parametric settings.

Model	Structural Units (~)	Trainable Params	Total Params	Total Params Size (MB)	Image.NetFile Size (MB)
VGG16	16	49.2 K	134 M	537.24	528.09
GoogLeNet	22	12.3 K	12.0 M	47.97	49.73
ResNet50	50	24.6 K	23.5 M	94.13	97.80
SqueezeNet_1_0	18	6.2 K	741 K	2.97	4.79
ResNeXt50_32x4d	50	24.6 K	23.0 M	92.02	96.56
Xception	36	24.6 K	20.8 M	83.33	87.45
MobileNet_V2	53	15.4 K	2.2 M	8.96	13.56
ShuffleNet_V2_X1_0	50	12.3 K	1.3 M	5.06	8.79
MNASNet_1_0	88	15.4 K	3.1 M	12.47	17.41
EfficientNet_B0	82	15.4 K	4.0 M	16.09	20.46
MobileNet_V3_Large	55	15.4 K	4.2 M	16.87	21.11
RegNet_X_800MF	16	8.1 K	6.6 M	26.38	26.20
RegNet_Y_800MF	16	9.4 K	5.7 M	22.63	24.77
ConvNeXt_V1	53	9.2 K	27.8 M	111.32	109.16

protocol using the same data preprocessing and augmentation steps. The Adam optimizer was employed with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 1 \times 10^{-8}$, along with a fixed learning rate of 0.0001 and a weight decay of 1×10^{-5}. No learning rate scheduler was applied to maintain consistent optimization across models. All newly added classification heads were initialized using PyTorch's default Kaiming uniform weight initialization method to ensure stable gradient propagation and convergence. An early stopping strategy monitored the validation loss with a patience value of 3 epochs, and model check-pointing was employed to preserve the best-performing weights. Each model was trained for a maximum of 10 epochs with a batch size of 32. During training, all convolutional layers were frozen and only the final classification head was retrained, as detailed in Table 2, which lists the total and trainable parameters for each model. All experiments were conducted with a fixed random seed (42) to ensure reproducibility across architectures (Abou Baker et al., 2022; Fotovvatikhah et al., 2025).

All CNN models used in this study were implemented with pre-trained weights on ImageNet via the TorchVision library and integrated using the PyTorch Lightning framework. The Adam optimizer with a constant learning rate was employed for training. Model checkpointing was activated in conjunction with early stopping to ensure that only the most effective version of each model was preserved. Training was conducted on a machine with an Intel Core i5 processor, 16 GB RAM, and Windows 10 (64-bit), using Python 3.12, PyTorch 2.3.1, TorchVision 0.18.1, and PyTorch Lightning 2.1.4. All experiments were conducted on a CPU environment to ensure fair and reproducible comparison across models without GPU acceleration bias. In addition, GPU-based or edge inference implementations can be carried out to evaluate model latency, throughput, and energy efficiency under realistic industrial conditions. Further optimization of these models for GPU acceleration and embedded hardware would enable their real-time deployment in practical recycling and waste management systems (Raschka et al., 2020; Fotovvatikhah et al., 2025).

Model performances were evaluated using common classification metrics, including accuracy, F1-score, precision, recall, and confusion matrix analysis (Gurcan and Soylu, 2024; Fotovvatikhah et al., 2025; Nahiduzzaman et al., 2025). Additionally, computational metrics such as training time and number of parameters were also considered to assess the resource efficiency of each model. This multidimensional evaluation framework enabled the identification of models that not only offer high classification accuracy but also meet the efficiency demands of real-time and mobile applications, such as waste classification.

Table 3
Performance comparison of 14 pre-trained CNN models based on classification metrics and training time (sec) on the waste classification task.

Model	Accuracy (Train)	Accuracy (Val)	Accuracy (Test)	Precision (Test)	Recall (Test)	F1-Score (Test)	Training Time
ConvNeXt_V1	0.9705	0.9639	0.9695	0.9560	0.9585	0.9572	10,010
ResNet50	0.9526	0.9428	0.9493	0.9234	0.9301	0.9264	15,631
VGG16	0.9650	0.9381	0.9472	0.9212	0.9230	0.9215	22,851
ResNeXt50_32x4d	0.9514	0.9424	0.9454	0.9211	0.9243	0.9222	19,948
MobileNet_V2	0.9450	0.9347	0.9390	0.9084	0.9125	0.9095	7219
MobileNet_V3_Large	0.9436	0.9338	0.9369	0.9056	0.9111	0.9080	5282
MNASNet_1_0	0.9405	0.9381	0.9369	0.9082	0.9104	0.9083	5904
RegNet_Y_800MF	0.9318	0.9248	0.9287	0.8968	0.9070	0.9003	6443
RegNet_X_800MF	0.9300	0.9248	0.9278	0.8979	0.9002	0.8978	6649
EfficientNet_B0	0.9276	0.9201	0.9227	0.8854	0.8981	0.8905	8312
SqueezeNet_1_0	0.9249	0.9042	0.9210	0.8845	0.8925	0.8878	5620
GoogLeNet	0.9140	0.9033	0.9072	0.8627	0.8636	0.8615	7749
Xception	0.9201	0.9072	0.9012	0.8504	0.8576	0.8525	22,123
ShuffleNet_V2_X1_0	0.8645	0.8724	0.8694	0.8162	0.8039	0.8061	2856
Mean	**0.9344**	**0.9250**	**0.9287**	**0.8956**	**0.8995**	**0.8964**	**10,471**

4. Experimental results

4.1. Classification performance and training efficiency of pre-trained CNN models

In this study, a total of 14 different pre-trained CNN models, each with distinct architectural and computational characteristics, were comprehensively analyzed for the task of waste image classification. The models were evaluated based on training, validation, and test accuracies, as well as precision, recall, F1-score (on the test set), and training time (in seconds). A detailed comparison of the models according to these metrics is presented in Table 3. In this table, the models are listed in descending order based on their test accuracy.

As can be seen from the table, the highest test accuracy was achieved by the ConvNeXt_V1 model with 96.95 %. This model, built on a modern convolutional architecture, demonstrated superior performance in both overall accuracy and F1-score. It was followed by ResNet50 (94.93 %), VGG16 (94.72 %), and ResNeXt50_32x4d (94.54 %). These high-performing models commonly feature residual connections and deep architectures, which are particularly effective in classification tasks. On the other hand, lightweight models such as MobileNet_V2 (93.90 %), MobileNet_V3_Large (93.69 %), and MNASNet_1_0 (93.69 %) also delivered competitive accuracy levels while standing out in terms of computational efficiency. Lower test accuracies were observed for ShuffleNet_V2_X1_0 (86.94 %), Xception (90.12 %), and GoogLeNet (90.72 %). These models typically have fewer parameters and shallower architectures, which may explain their relatively lower performance.

Beyond accuracy, the models were also compared in terms of training time. VGG16 (22,851 s), Xception (22,123 s), and ResNeXt50_32x4d (19,948 s) were the models with the longest training durations. In contrast, ShuffleNet_V2_X1_0 (2856 s), MobileNet_V3_Large (5282 s), and SqueezeNet_1_0 (5620 s) were trained in much shorter times, offering better computational efficiency. While there is no direct correlation between training time and accuracy, certain models achieving acceptable performance in shorter durations make them attractive for resource-constrained systems.

To provide a holistic view of classification performance and resource efficiency, two key metrics, test accuracy and training time, are also presented in Fig. 4 and Fig. 5. These visualizations help highlight the trade-offs between accuracy and computational cost, offering an intuitive and comparative perspective on the suitability of each model for different scenarios. When both indicators are considered together, models that balance accuracy and efficiency, such as MobileNet_V3_Large and MNASNet_1_0, are more suitable for practical applications.

Specifically, MobileNet_V3_Large, MNASNet_1_0, and MobileNet_V2 models stand out by delivering balanced performance in terms of accuracy (93.69 %–93.90 %) and training time (5282–7219 s). In contrast, RegNet_Y_800MF and RegNet_X_800MF provide lower accuracies (92.87 %–92.78 %) and require longer training times, making them less favorable in terms of efficiency. Although ShuffleNet_V2_X1_0 achieved the lowest test accuracy (86.94 %), it trained in the shortest time (2856 s), indicating a trade-off. However, in high-stakes applications where accuracy is a priority, deeper models such as ConvNeXt_V1, despite their longer training durations, may offer a clear advantage. Therefore, the selection of a classifier model depends on whether the application prioritizes accuracy or computational efficiency and training time.

To further examine an optimal balance between predictive performance and computational efficiency in the waste classification

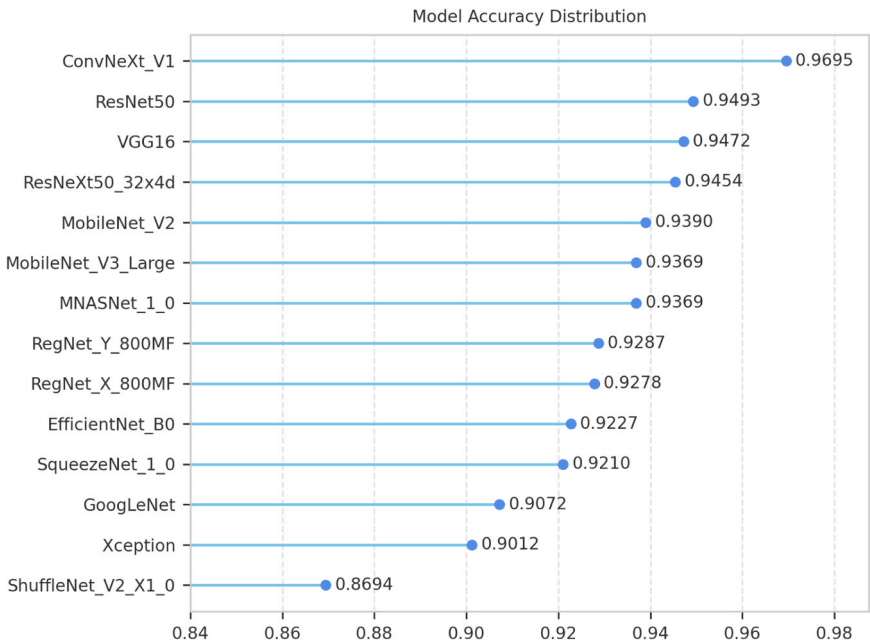


Fig. 4. Comparison of classification accuracies for pre-trained CNN models.

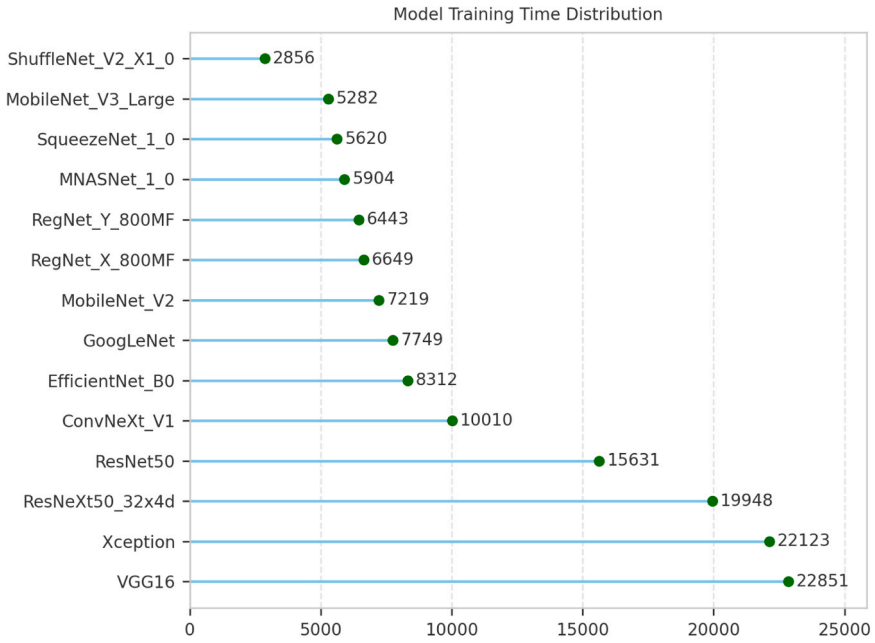


Fig. 5. Comparison of training times (in seconds) for pre-trained CNN models.

task, a Pareto frontier analysis was conducted using the test accuracy and training time of all 14 pre-trained CNN models (see Fig. 6). Each point on the chart represents one model, with the Pareto-optimal frontier (green dashed line) denoting architectures that offer the best trade-offs between accuracy and efficiency. The results demonstrate that ConvNeXt_V1 achieved the highest test accuracy (96.95 %) but required the longest training time due to its deeper architecture and larger parameter count. In contrast, lightweight models such as MobileNet_V2, MobileNet_V3_Large, and ShuffleNet_V2_X1_0 delivered competitive accuracies (above 93 %) with substantially reduced computational cost. These models lie on the Pareto frontier, confirming their strong efficiency–performance trade-off.

The implications of our findings are particularly relevant for real-world waste management systems, where speed, energy efficiency, and deployability on low-power devices are essential. Hence, lightweight CNNs provide an optimal balance between accuracy and efficiency for embedded, real-time waste classification. As a result, this comprehensive evaluation of 14 pre-trained CNN models highlights how different architectures trade off predictive performance and computational efficiency, offering valuable insight into their suitability for various deployment scenarios. Building on these findings, the next section presents a detailed class-wise performance and prediction error analysis to further examine the learning behavior of the top-performing models.

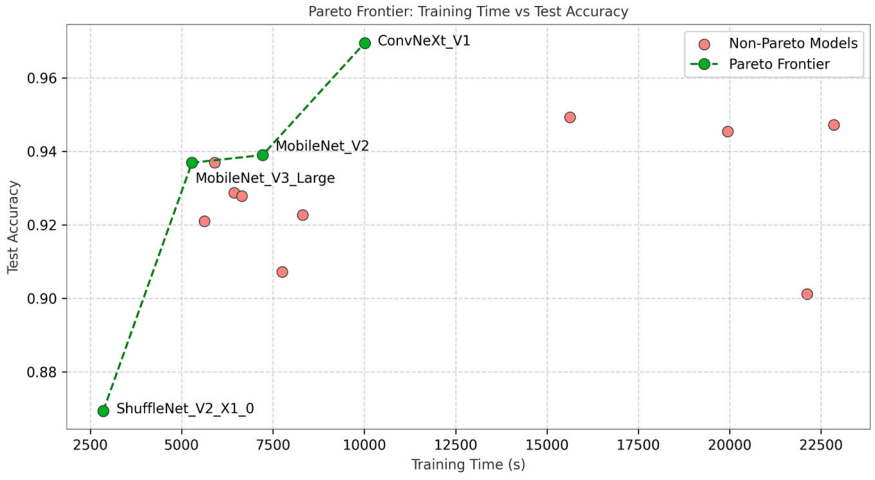


Fig. 6. Pareto frontier of pre-trained CNN models based on training time and test accuracy.

4.2. Class-wise performance and prediction error analysis

Class-wise performance and error analysis goes beyond evaluating machine learning models solely based on global performance metrics such as overall accuracy, by revealing each model's strengths and weaknesses on a per-class basis. Such analysis is particularly valuable for datasets with class imbalance, as it allows for a more detailed assessment of the model's sensitivity (recall), specificity (precision), and discriminative ability for each class. In this context, a detailed per-class evaluation was conducted for the three models that achieved the highest overall test accuracies: ConvNeXt_V1, ResNet50, and VGG16. Based on the F1-score values of each model, this analysis compares how accurately each model classified individual waste categories. Table 4 presents the class-wise F1-scores of the three models (ConvNeXt_V1, ResNet50, and VGG16) along with the corresponding training, validation, and test sample counts, total class sizes, and mean F1 values. The classes are sorted in descending order according to the average F1-score in the final column. To more clearly illustrate performance differences across classes, the mean F1-scores for each class are also visualized in Fig. 7. As seen in the figure, most classes achieved mean F1-scores above 0.94; however, the plastic (0.8490), white-glass (0.8725), and metal (0.8773) classes exhibited a significant drop in performance.

The results suggest that, in addition to class sample sizes, qualitative factors such as visual distinctiveness, color similarity, and texture overlap also play a decisive role in model performance. For example, high classification performance was achieved in classes with large sample sizes such as clothes (794, F1: 0.9898) and shoes (305, F1: 0.9768), indicating that sufficient data volume enhances the model's generalization capacity. On the other hand, some classes with relatively fewer samples, such as trash, green-glass, cardboard, and brown-glass, still achieved high F1-scores, indicating that the model generalized effectively for these categories. Nevertheless, despite similar sample sizes, there were notable performance disparities between some classes. In particular, the low F1-scores observed in the metal (110), white-glass (104), and plastic (120) classes (all below 88 %) indicate that the model struggled to distinguish between these visually similar categories, which likely share structural properties and reflective/translucent surface characteristics.

A general review of Table 4 shows that ConvNeXt_V1 achieved the highest and most consistent F1-scores across classes. While ResNet50 performed comparably in most categories, VGG16, although generally yielding lower scores, delivered competitive results in classes such as clothes, shoes, and biological. Across all three models, the plastic class exhibited the lowest F1-scores and was the most challenging to classify.

Following this evaluation, a complementary analysis was conducted to quantify how class weighting influenced recall performance across different categories. To achieve this, balanced class weights were incorporated into the CrossEntropyLoss function, assigning higher weights to minority classes to ensure fair learning across all categories. Given the noticeable class imbalance in the dataset, a potential decline in learning performance was expected for underrepresented categories. Therefore, recall values (Δ Recall) were compared before and after applying class weighting to determine whether this approach enhances the model's sensitivity toward minority classes and mitigates the effects of data imbalance.

Table 5 presents the total number and percentage of samples for each class, along with the recall improvements (Δ Recall) achieved by the three CNN models (ConvNeXt_V1, ResNet50, and VGG16) after applying class weighting, as well as the mean Δ Recall across all models. The classes are sorted in descending order according to their total number of samples, allowing a clearer observation of the dataset imbalance and its influence on model performance.

The analysis results show that in classes with fewer samples (such as white-glass, metal, brown-glass, and green-glass), class weighting led to recall improvements ranging from 1.6 % to 2.3 %. In particular, recall increased by + 2.25 % for white-glass, + 2.12 % for metal, + 1.75 % for brown-glass, and + 1.68 % for green-glass. These gains demonstrate that class weighting provided a clear advantage for classes with relatively fewer samples. In contrast, performance in majority classes with larger sample sizes (such as clothes and shoes) remained mostly stable, with Δ Recall values fluctuating within the ±0.01 range. Across all classes and models, the average Δ Recall was calculated as + 0.4 %. These findings indicate that the class weighting strategy effectively reduces bias arising from data imbalance, enhances the model's sensitivity to underrepresented categories, and maintains overall performance stability.

Table 4
Class-wise F1-score comparison of the top three performing models.

Class	Train Set (70 %)	Val. Set (15 %)	Test Set (15 %)	TotalSize (#)	TotalSize (%)	Conv NeXt_V1	ResNet50	VGG16	Mean F1
clothes	3682	849	794	5325	34.32	0.9911	0.9885	0.9898	0.9898
shoes	1388	284	305	1977	12.74	0.9838	0.9724	0.9741	0.9768
biological	696	164	125	985	6.35	0.9880	0.9494	0.9558	0.9644
battery	638	141	166	945	6.09	0.9674	0.9550	0.9521	0.9582
trash	501	87	109	697	4.49	0.9862	0.9541	0.9170	0.9524
green-glass	419	91	119	629	4.05	0.9705	0.9496	0.9231	0.9477
cardboard	646	124	121	891	5.74	0.9633	0.9312	0.9456	0.9467
brown-glass	414	98	95	607	3.91	0.9684	0.9319	0.9305	0.9436
paper	744	146	160	1050	6.77	0.9527	0.9367	0.9362	0.9419
metal	538	121	110	769	4.96	0.9091	0.8734	0.8493	0.8773
white-glass	571	100	104	775	5.00	0.9135	0.8442	0.8598	0.8725
plastic	623	122	120	865	5.58	0.8926	0.8299	0.8246	0.8490

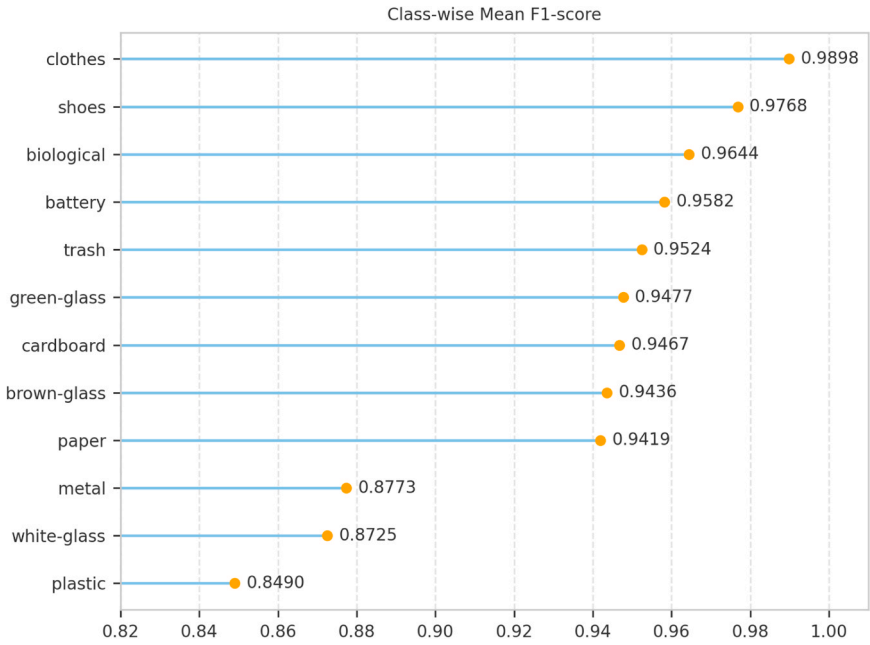


Fig. 7. Comparison of class-wise mean F1-scores of the top three performing models.

Table 5
Class-wise recall improvement (Δ Recall) using class weighting for the top three CNN models.

Class	Total Size (#)	Total Size (%)	ConvNeXt_V1 (Δ Recall)	ResNet50 (Δ Recall)	VGG16 (Δ Recall)	Mean (Δ Recall)
Clothes	5325	34.32	−0.0038	−0.0114	−0.0063	−0.0072
Shoes	1977	12.74	−0.0033	−0.0066	−0.0065	−0.0055
Paper	1050	6.77	0.0000	−0.0063	0.0000	−0.0021
Biological	985	6.35	0.0000	−0.0080	−0.0080	−0.0053
Battery	945	6.09	−0.0061	−0.0061	0.0000	−0.0041
Cardboard	891	5.74	0.0083	0.0000	0.0083	0.0055
Plastic	865	5.58	0.0000	−0.0084	−0.0250	−0.0111
White-glass	775	5.00	0.0097	0.0481	0.0096	0.0225
Metal	769	4.96	0.0000	0.0455	0.0182	0.0212
Trash	697	4.49	0.0000	0.0000	0.0092	0.0031
Green-glass	629	4.05	0.0084	0.0252	0.0168	0.0168
Brown-glass	607	3.91	0.0105	0.0315	0.0105	0.0175
Total	**15515**	**100**	**0.0020**	**0.0087**	**0.0023**	**0.0043**

4.3. Confusion matrices and classification reports of the top performing models

Following this general comparison, confusion matrices and classification reports were provided for each of the top three models in terms of test accuracy: ConvNeXt_V1, ResNet50, and VGG16. These visualizations enable a detailed evaluation of both class-level performance and the learning dynamics of each model. The class-wise performance of ConvNeXt_V1, the highest-performing model, is presented both visually and quantitatively in Fig. 8. The confusion matrix in Fig. 8A is structured such that rows represent the actual classes, while columns represent the model's predicted classes. As illustrated, the model performed nearly flawlessly in classifying clothes, biological, and trash samples. While it achieved generally high accuracy in other classes as well, limited misclassifications were observed in plastic, metal, and white-glass classes. The most frequent confusions occurred between the following class pairs: clothes–shoes (7 samples), paper–cardboard (5 samples), plastic–metal (5 samples), and white-glass–plastic (5 samples). The classification report heatmap in Fig. 8B visualizes the precision, recall, and F1-score values obtained by ConvNeXt_V1 for each class using color intensity.

Figs. 9A and 9B present the class-wise evaluation of ResNet50, the second-best model in terms of test accuracy. The confusion matrix in Fig. 8A is similarly structured, with actual classes along the rows and predicted classes along the columns. Overall, the model achieved high classification accuracy, but notable misclassifications were observed in certain classes. Specifically, 10 samples from the white-glass class were incorrectly classified as paper. Likewise, 8 paper samples were predicted as cardboard, and 11 clothes samples were misclassified as paper. The metal class was confused with battery (4 samples), paper (2 samples), and plastic (3 samples). Additionally, several plastic samples were misclassified as metal, paper, or white-glass. These patterns indicate that the model

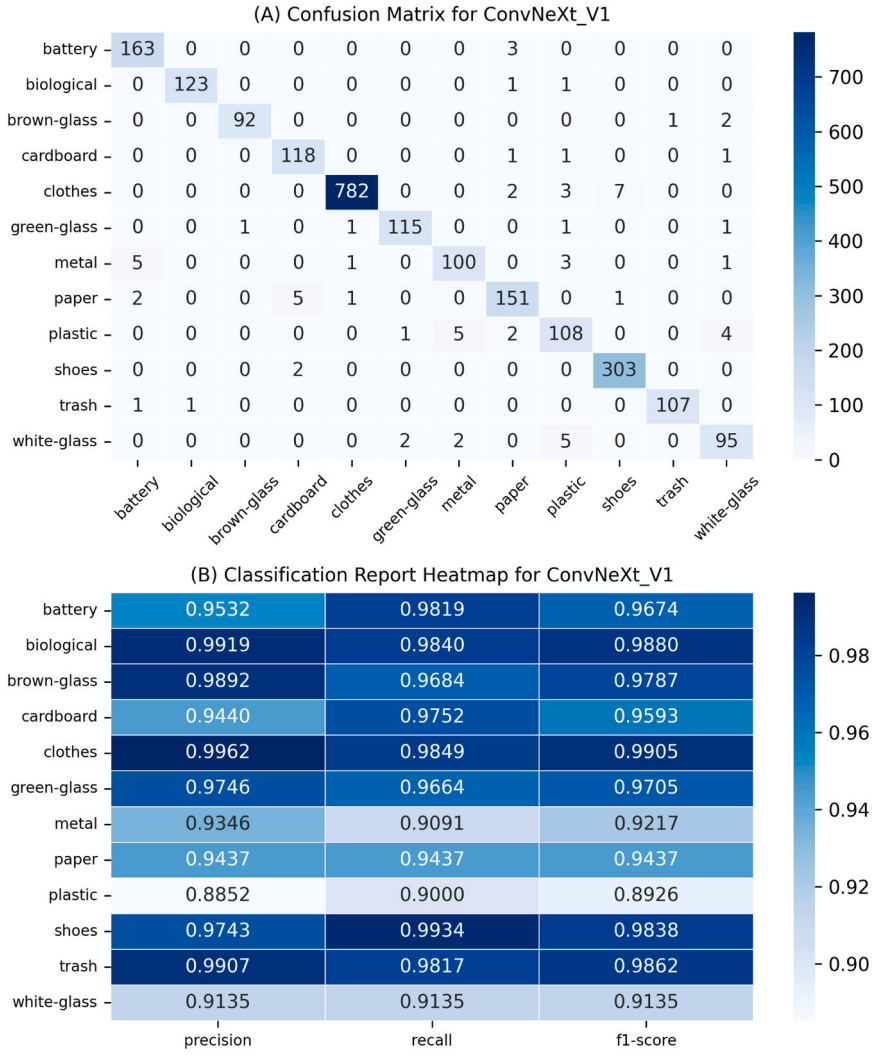


Fig. 8. Class-wise evaluation of the ConvNeXt_V1 model based on confusion matrix and metric heatmap.

struggled to distinguish between certain visually and structurally similar classes. The heatmap in Fig. 8B presents precision, recall, and F1-score values per class through color intensity, thereby visualizing ResNet50's overall per-class performance.

Fig. 10A and B display the class-wise evaluation of VGG16, which ranked third in test accuracy. The confusion matrix in Fig. 9A also follows the same format, with rows representing true classes and columns showing predictions. The matrix reveals that the model achieved high accuracy in shoes (301/305), clothes (780/794), trash (105/109), and paper (154/160) classes. However, notable confusion was also observed between some classes. In particular, the plastic class was frequently misclassified as white-glass (10 samples), trash (7 samples), and paper (5 samples). The metal class was confused with battery (4 samples), paper (4 samples), and plastic (3 samples). Similarly, the white-glass class was misclassified as plastic (5 samples) and metal (3 samples). The heatmap in Fig. 9B shows the precision, recall, and F1-score values calculated by the VGG16 model for each class using color intensity. It clearly highlights which classes were classified accurately and which ones were more challenging for the model.

4.4. Misclassification patterns and explainable AI (XAI) insights

This section presents an in-depth analysis of misclassification behavior and interpretability findings derived from the deep learning models. Although the 14 pre-trained CNN architectures achieved high overall accuracy, all models exhibited certain confusion tendencies among specific classes. In particular, the low F1-scores observed for the plastic, metal, and white-glass categories indicate that the models struggled to distinguish their visual similarities. This analysis examined the underlying causes of these confusions and interpreted the visual features guiding the models' predictions, focusing on the best-performing model, ConvNeXt_V1.

The analysis revealed that plastic, metal, and white-glass objects share highly similar surface characteristics. These materials often exhibit glossy, reflective, or semi-transparent textures, producing overlapping visual cues under varying lighting conditions or

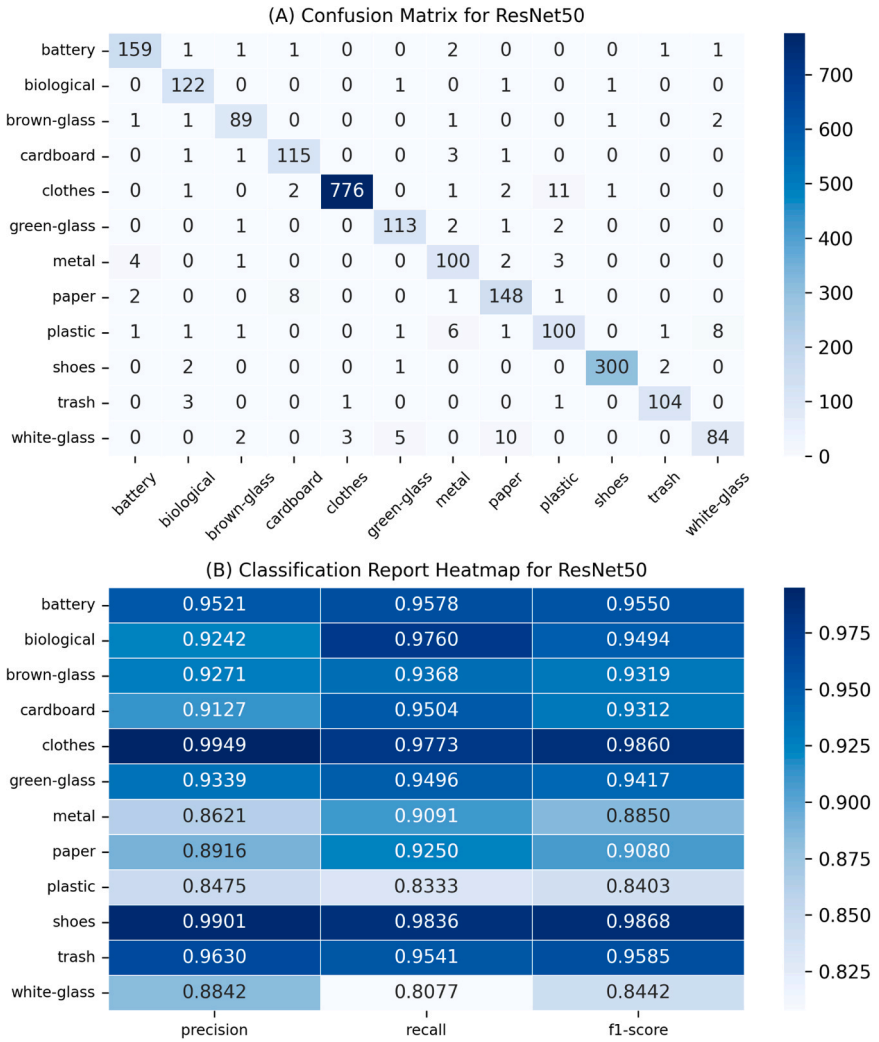


Fig. 9. Class-wise evaluation of the ResNet50 model based on confusion matrix and metric heatmap.

complex backgrounds. Such similarities lead to blurred class boundaries in the model's feature space, resulting in mutual misclassifications. As shown in Table 4, their relatively lower F1-scores quantitatively reflect this visual overlap. Fig. 11 presents misclassified samples from these categories, visually demonstrating how the model confuses objects with similar color tones and surface reflections.

To gain a deeper understanding of the model's decision process, Grad-CAM-based attention maps were generated for ConvNeXt_V1. Grad-CAM (Gradient-weighted Class Activation Mapping) is an explainable AI (XAI) technique that highlights the image regions most influential in a model's prediction, thereby visualizing the focus areas driving classification. The attention maps in Fig. 12 show that, while high-performing classes (e.g., clothes, shoes, biological) direct attention across the entire object, low-performing classes (plastic, metal, white-glass) tend to focus on background or reflection regions. The model frequently attends to light reflections on plastic bottles or shiny metallic surfaces, indicating a strong sensitivity to material-based brightness variations that obscure object boundaries.

As a result, this XAI-based analysis demonstrates that even high-performing CNNs encounter difficulties in distinguishing visually similar materials. The interpretability results enhance the transparency of model behavior and provide practical insights for reducing confusion through techniques such as class-aware data augmentation, attention refinement, and contrastive learning.

4.5. *Statistical significance analysis*

To statistically validate the performance differences among the pre-trained CNN models, the non-parametric Friedman test was employed. This test was selected because it does not assume a normal distribution of data and is particularly suitable for comparing three or more models evaluated under identical experimental conditions. In contrast, a paired *t*-test is limited to pairwise comparisons and requires the normality assumption, which may not hold in this case. The Friedman test was therefore considered more appropriate

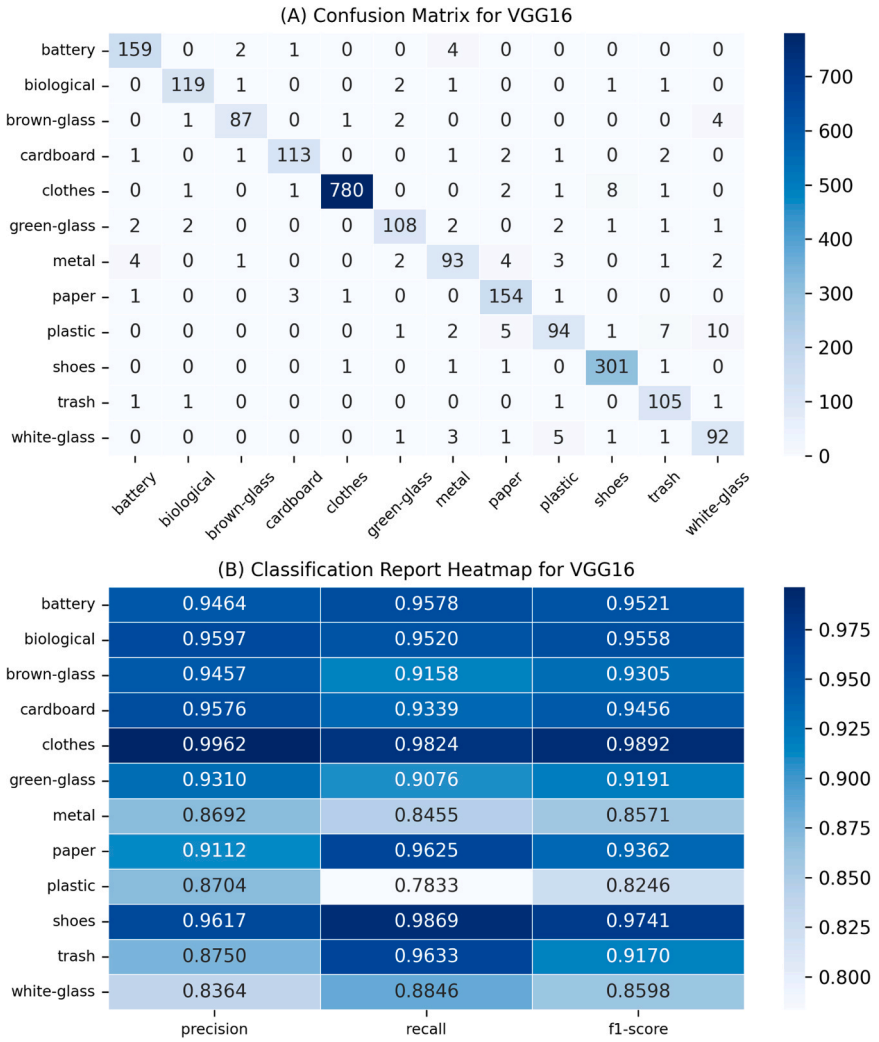


Fig. 10. Class-wise evaluation of the VGG16 model based on confusion matrix and metric heatmap.

to evaluate multiple models simultaneously while avoiding the accumulation of Type I errors that can occur with repeated *t*-tests.

The analysis was performed using test accuracy values obtained from three independent runs with different random seeds (42, 56, and 99). The results of the Friedman test revealed statistically significant differences among the models ($\chi^2 = 38.943$, $p = 0.000204$, $p < 0.05$), confirming that the observed performance variations were not due to random chance. This finding indicates that ConvNeXt_V1 achieved statistically higher accuracy compared to lightweight architectures such as MobileNet_V3_Large, MNASNet_1_0, and SqueezeNet_1_0. Hence, the proposed model's superior performance is statistically supported in addition to its numerical advantage.

5. Discussion

This study provided a comprehensive performance comparison of 14 pre-trained CNN models with different architectural structures under the same dataset and training protocol. In particular, detailed class-wise evaluations enabled in-depth insights into model performance beyond overall accuracy scores. Results were evaluated based on accuracy, class-level performance, visual confusion, and computational efficiency.

A review of the existing literature reveals that numerous CNN-based approaches have been proposed for the waste classification problem (Abou Baker et al., 2022; Wu et al., 2023; Fotovvatikhah et al., 2025). Prominent studies in the field, along with the methods and findings they reported, are summarized comparatively in Table 6. As seen in the table, existing approaches generally fall into three major categories. The first group consists of deep learning models trained from scratch, where the architecture is designed and trained directly on the task-specific dataset (Bobulski and Kubanek, 2021; Chhabra et al., 2024; Wu et al., 2024). The second group comprises transfer learning (TL) approaches, where pre-trained models are restructured and fine-tuned for the target domain (Faria et al., 2021;

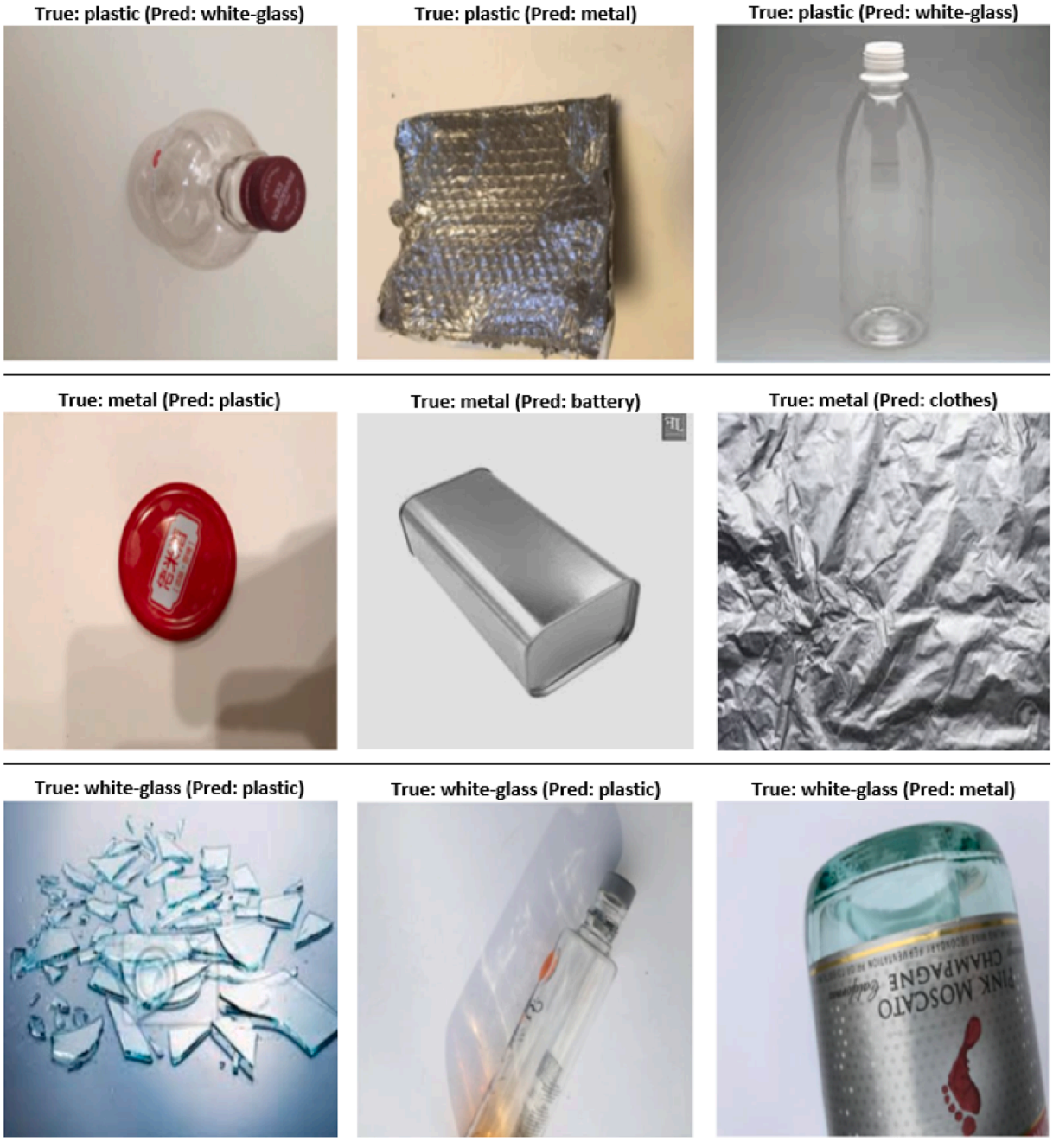


Fig. 11. Misclassified samples between visually similar categories (plastic, metal, and white-glass).

Komathi et al., 2023; Malik et al., 2022; Poudel and Poudyal, 2022). The third group includes hybrid systems that combine deep features with traditional classifiers such as SVM or Random Forest (Jain et al., 2022; Al Duhayyim et al., 2023; Mishra et al., 2024). Beyond these waste-specific studies, recent literature across multiple application domains shows a growing emphasis on lightweight and explainable deep learning architectures (Kaya and Gürsoy, 2023; Gurcan and Soylu, 2024; Ahamed et al., 2024a; Gurcan, 2025; Nahiduzzaman et al., 2025; Naznine et al., 2025).

In recent years, explainable, lightweight, and high-accuracy deep learning architectures have been proposed across various domains. These studies have focused on developing hybrid frameworks that integrate deep feature extraction with conventional learning strategies for applications such as medical diagnosis, agricultural disease detection, industrial quality control, environmental monitoring, and waste classification (Kaya and Gürsoy, 2023; Wu et al., 2023; Arbeláez-Estrada et al., 2023; Ahamed et al., 2024b, 2025;

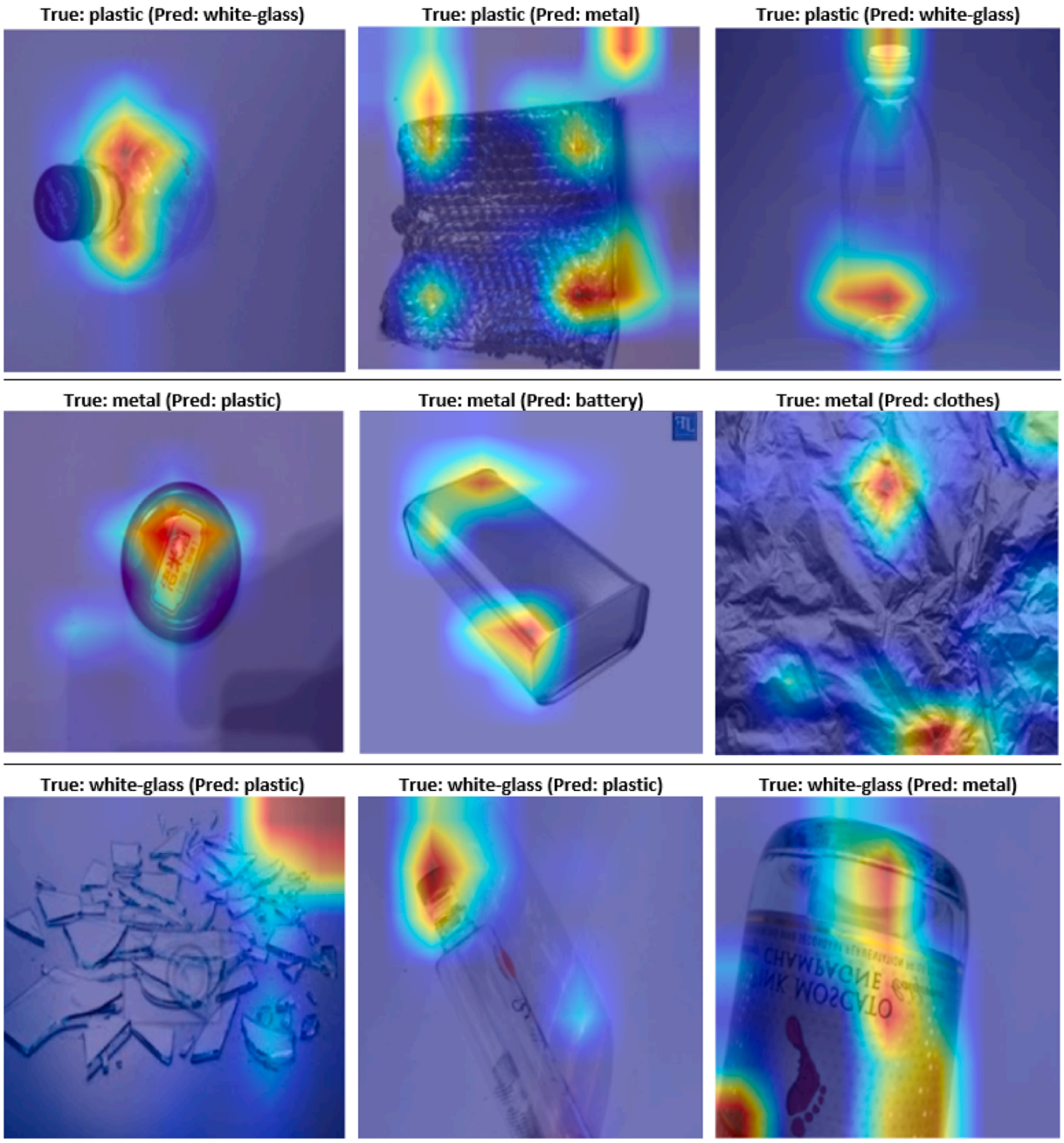


Fig. 12. Grad-CAM visualization of misclassified samples between visually similar categories (plastic, metal, and white-glass).

Gürsoy and Kaya, 2025; Kilinc et al., 2025; Madhavi et al., 2025; Yevle and Mann, 2025; Fotovvatikhah et al., 2025). The main goal of these approaches has been to reduce model complexity and computational cost while maintaining high predictive accuracy. Many works have introduced compact variants of classical CNN architectures or incorporated attention and sequential components to enhance contextual awareness and generalization (Huang et al., 2021; Adami and Schiavon, 2021; Arbeláez-Estrada et al., 2023; Naznine et al., 2025). More recent hybrid frameworks have combined multi-level feature representations to further improve accuracy, often at the expense of increased computational demand and parameter size. Parallel to these advancements, explainable artificial intelligence (XAI) methods have become increasingly prominent to visualize model decisions and strengthen interpretability and trustworthiness (Chhabra et al., 2024; Ahamed et al., 2024a; Disci et al., 2025; Kilinc et al., 2025; Madhavi et al., 2025).

Unlike previous research, this study evaluates 14 pre-trained CNN models under identical conditions, comparing accuracy, class-level performance, and efficiency. This research provides a unified transfer learning-based comparison of models using different pre-

Table 6
Comparison of the proposed models and previous studies on waste classification.

Year	Study	Algorithm	Class Count	ACC
2020	(Huang et al., 2020)	Pre-trained CNN	6	96.50
2021	(Bian et al., 2021)	SSD-Mobilenet	7	98.50
2021	(Zhang et al., 2021)	DenseNet 169	5	82.80
2021	(Bobulski and Kubanek, 2021)	Custom CNN	4	96.41
2021	(Faria et al., 2021)	VGG16	4	88.42
2021	(Kumar et al., 2021)	SVM	4	96.50
2021	(Zheng and Gu, 2021)	Ensemble CNN	6	94.55
2021	(Huang et al., 2021)	VIT	6	96.48
2022	(Altikat et al., 2022)	Deep CNN	4	83.00
2022	(Malik et al., 2022)	EfficientNet-B0	10	85.00
2022	(Jain et al., 2022)	VGG16 +Custom CNN	12	96.00
2022	(Poudel and Poudyal, 2022)	InceptionResNetV2	7	93.82
2023	(Konstantinidis et al., 2023)	YoloV8	2	96.00
2023	(Komathi et al., 2023)	Pre-trained CNN	6	92.60
2023	(Al Duhayyim et al., 2023)	SVM+ELM+GBT	2	98.85
2023	(Yong et al., 2023)	MobileNetV2	4	82.92
2023	(Li and Chen, 2023)	CNN+Graph-LSTM	6	97.50
2024	(Lilhore et al., 2024)	Hybrid CNN-LSTM	2	95.45
2024	(Chhabra et al., 2024)	Custom Deep CNN	2	93.28
2024	(Pitakaso et al., 2024)	Ensemble CNN	12	94.05
2024	(Subashini and Vignesh, 2024)	MLP-CNN	7	92.66
2024	(Mishra et al., 2024)	DenseNet-121 + SVM	6	99.84
2024	(Prova, 2024)	VIT	12	95.00
2024	(Wu et al., 2024)	Custom CNN	12	99.41
2025	(Ranjbar et al., 2025)	MobileNetV3	4	91.04
2025	(Oza et al., 2025)	Pre-trained CNN	10	99.52
2025	(Nahiduzzaman et al., 2025)	DP-CNN+En-ELM	36	96.00
2025	(Madhavi et al., 2025)	Swin+ConvNeXt	12	98.97
2025	(Islam et al., 2025a)	ECCDN-Net	2	96.10
2025	This study*	ConvNeXt_V1 (Pre-trained CNN)	12	96.95

* This study implemented 14 pre-trained CNN models. The best-performing model, ConvNeXt_V1, is reported here.

training strategies under consistent conditions. ConvNeXt_V1 exhibited the best performance (96.95 % accuracy, 95.72 % F1-score), demonstrating strong competitiveness with existing approaches. This strong performance can be attributed to architectural enhancements inspired by Transformer principles, such as large kernels, LayerNorm, and GELU activations (Liu et al., 2022). While classical models such as ResNet50 and VGG16 offer high accuracy, they are more computationally expensive and less efficient in terms of training time and parameter usage (Faria et al., 2021; Islam et al., 2025a). Lightweight models (MobileNetV3-Large, MNASNet_1_0, MobileNetV2) are more efficient in training time and resource use, though slightly less accurate (Yong et al., 2023; Ranjbar et al., 2025; Islam et al., 2025a). In particular, ShuffleNet_V2_X1_0 and MobileNet models achieved good accuracy with very short training, making them suitable for real-time use in limited-resource settings (Khan et al., 2020; Abou Baker et al., 2022; Zhang et al., 2025).

The class-wise performance analysis revealed that, beyond model architecture, visual diversity within the dataset and class separability also play a critical role in determining classification performance. Specifically, all models performed well in classes such as clothes, shoes, and biological, which had high sample counts and distinctive visual features. In contrast, significant drops in F1-score values were observed for classes like plastic, metal, and white-glass. These categories were among the most frequently misclassified, primarily due to similar surface characteristics (e.g., reflectivity, translucency), as well as color and texture resemblance. Notably, the plastic–metal–white-glass class trio formed the most prominent cluster where inter-class boundaries became blurred. This outcome indicates that not only the number of samples, but also data quality and intra-class consistency are critical for model performance (Jain et al., 2022; Pitakaso et al., 2024, p. 12; Prova, 2024; Wu et al., 2024; Ranjbar et al., 2025).

For the best-performing model, ConvNeXt_V1, the merits and demerits are presented as follows. Among the 14 pre-trained CNN architectures compared in this study, ConvNeXt_V1 achieved the highest test accuracy (96.95 %) and F1-score (0.9572), demonstrating the best overall performance. The main advantage of the model is its ability to reach a high level of accuracy by retraining only the classification head, which effectively reduces both training time and computational cost. The incorporation of large convolutional kernels, Layer Normalization, and GELU activation, inspired by Transformer architectures, enabled the model to produce more stable results, particularly for classes with fewer samples (Liu et al., 2022; Madhavi et al., 2025; Zhang et al., 2025).

Furthermore, the close alignment between accuracy, precision, and recall values indicates a strong generalization capability. However, some limitations still exist. Although the training time of ConvNeXt_V1 (10,010 s) is slightly below the average (10,471 s), it still requires higher GPU memory and computational resources compared to lightweight models such as MobileNet_V3_Large (5282 s) and MNASNet_1_0 (5904 s). The total of 27.8 million parameters may restrict real-time applications on embedded or edge devices with limited hardware capacity. The frozen transfer learning strategy used in this study achieved high accuracy with low computational cost. Future work will explore partial fine-tuning of upper layers to improve domain adaptation, while carefully balancing performance and efficiency. This approach may further enhance feature representation for complex waste categories (Huang et al., 2020; Zhang et al., 2021; Liu et al., 2022; Madhavi et al., 2025; Oza et al., 2025).

This study provides both theoretical contributions to the field of deep learning-based waste classification and practical implications for energy-efficient artificial intelligence applications. From a theoretical perspective, the comparative evaluation of 14 pre-trained CNN architectures validates the effectiveness of the transfer learning strategy. The results demonstrate that high accuracy can be achieved by retraining only the classification head, indicating that frozen feature extraction layers are sufficient to capture the general visual patterns of waste images. Moreover, the relationship between architectural complexity (total parameter count) and accuracy has been systematically examined.

From a practical standpoint, lightweight models such as MobileNet_V3_Large and MNASNet_1_0 offer short training times and fewer parameters, making them suitable for embedded systems and real-time waste classification on edge devices. In contrast, more complex architectures like ConvNeXt_V1 deliver higher accuracy but require greater GPU memory and computational resources. This comparison highlights that the balance between performance and hardware cost is a critical factor in the design of sustainable AI systems. The findings also provide guidance for several real-world applications. Lightweight CNN models can be deployed in smart recycling bins, waste-sorting robots, offline mobile classification applications, and automated inspection systems in waste collection facilities. Furthermore, the analysis of model efficiency and energy consumption offers a technical foundation for developing green computing-oriented embedded AI solutions. In this regard, the study serves as a concrete reference framework for future research focused on environmental sustainability and hardware optimization.

Although the findings of this study are supported by comprehensive model comparisons and class-level analyses, several limitations should be acknowledged. First, all experiments were conducted solely in a CPU environment, and no GPU-based training or speed evaluation was performed. As a result, the full computational potential of some models may not have been fully realized. The dataset was limited in size and scope, focusing on a specific waste context. Model generalizability was not tested on data from other regions or imaging conditions. While class imbalance was partially mitigated through a weighted loss function, no advanced techniques such as data augmentation or resampling were employed. Model evaluation focused only on classification metrics; qualitative aspects like explainability or user confidence were outside this study's scope.

For future work, strategies such as contrastive learning, attention-based mechanisms, and class-aware augmentation should be explored. These methods can help models better distinguish between visually similar and internally diverse classes. Model evaluation should also go beyond classification accuracy. Additional metrics, including energy consumption, computation time, and memory usage, need to be considered. Such metrics are essential for a more comprehensive assessment of model suitability in real-world applications. In practical use cases, such as smart waste sorting systems, mobile recycling assistants, and low-power embedded devices, choosing models that balance accuracy and resource efficiency is very important. In conclusion, the findings highlight that balancing predictive performance with computational efficiency is the key to building sustainable and deployable AI solutions for real-world waste management.

6. Conclusion

This study conducted a comparative evaluation of 14 different pre-trained CNN models under identical experimental settings for the task of automatic waste classification involving 12 waste categories using a publicly available dataset. The highest test accuracy (96.95 %) and F1-score (95.72 %) were achieved by the ConvNeXt_V1 architecture, demonstrating the effectiveness of modern convolutional models enhanced with Transformer-based architectural improvements. Beyond overall accuracy, class-wise analyses revealed that visual similarity among categories such as plastic, metal, and white glass negatively impacted classification performance. While deeper architectures (e.g., ResNet50, VGG16) achieved high accuracy, they incurred higher training times and computational costs. In contrast, lightweight models such as MobileNetV3-Large and MNASNet_1_0 offered reasonable accuracy with significantly shorter training durations, making them suitable options for resource-constrained systems.

Future work should focus on integrating advanced learning strategies (e.g., contrastive learning, attention mechanisms) and multi-dimensional evaluation metrics (e.g., explainability, energy efficiency, inference time) to improve model robustness and deployment in real-world waste classification systems. In conclusion, this study offers a systematic comparison of different CNN architectures in terms of accuracy, resource efficiency, and class-wise performance, and provides a holistic framework that can guide model selection for intelligent waste management systems.

CRediT authorship contribution statement

Fatih Gurcan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmet Soylu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

influence the work reported in this paper.

Data availability

The dataset is publicly available and cited in the article.

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